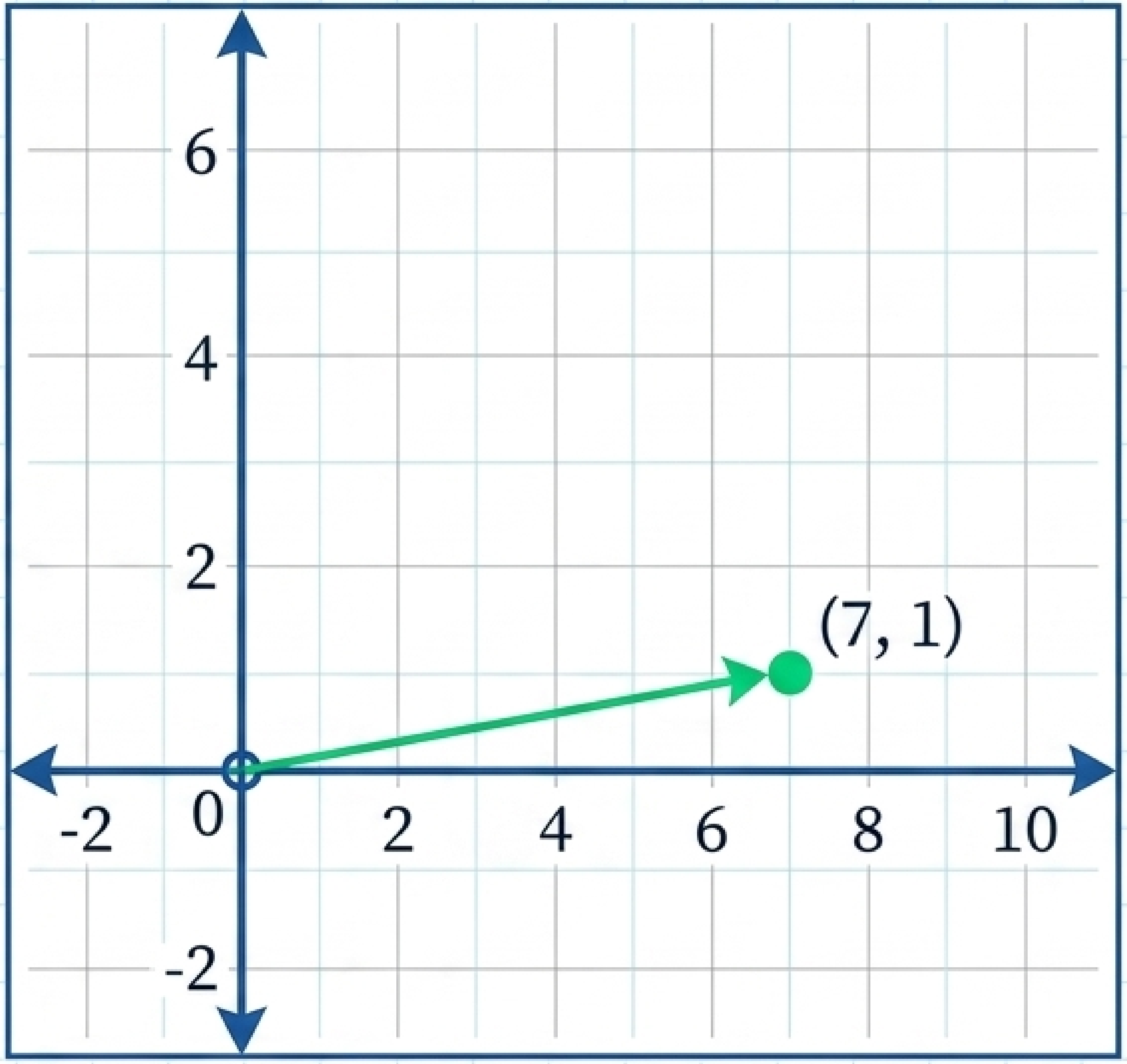
Data as Geometry

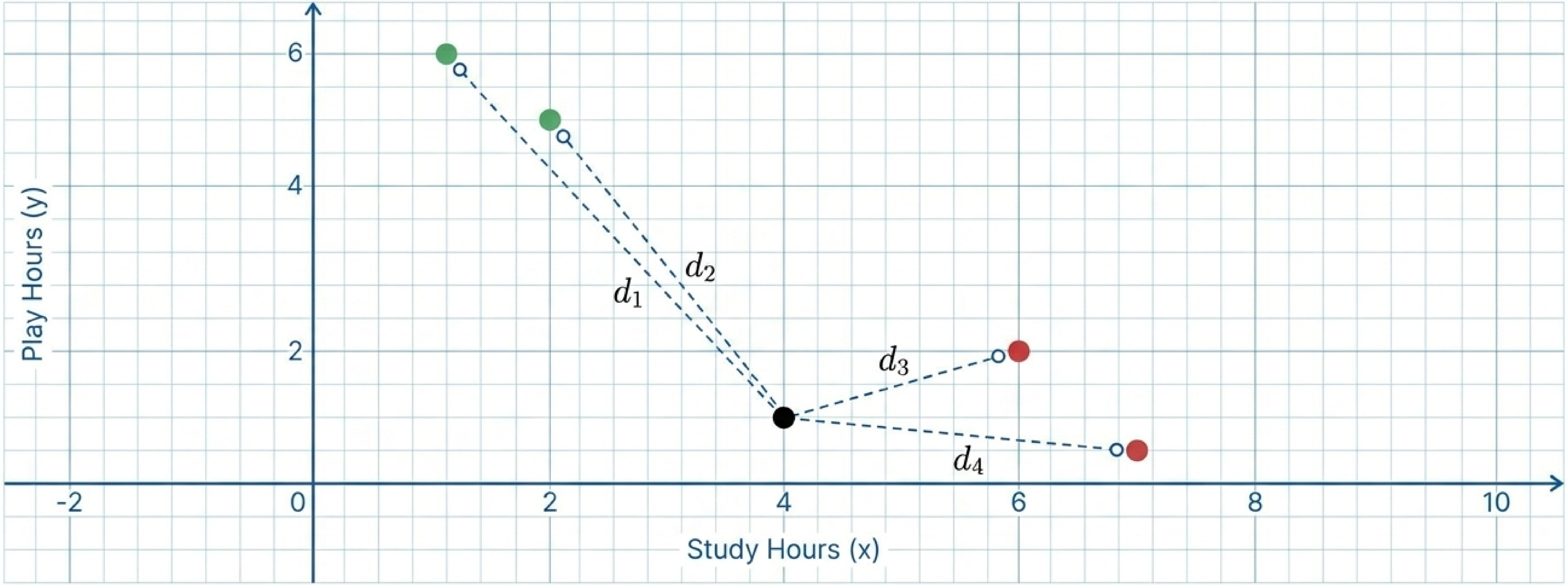
The linear algebra engine powering machine learning models.

A Visual Reference Guide

Study Hours (x)	Play Hours (y)	Result
7	1	Pass



In machine learning, every single row of data is translated into a vector.
The dataset is not a spreadsheet; it is a physical map of points.



New Unseen Student: Study 4, Play 1. Will they pass or fail?

The algorithm calculates the physical Euclidean distance between the new vector and all known vectors. Because $(4{,}1)$ is physically closer to the 'Pass' vectors, the model predicts a Pass.

The Scalar

$$\boxed{30}$$

Magnitude only (size/amount). No directional component.

Real-World Example:
Temperature, length of a hill (30 meters).

ML Application: Mean Centering (Shifting entire datasets).

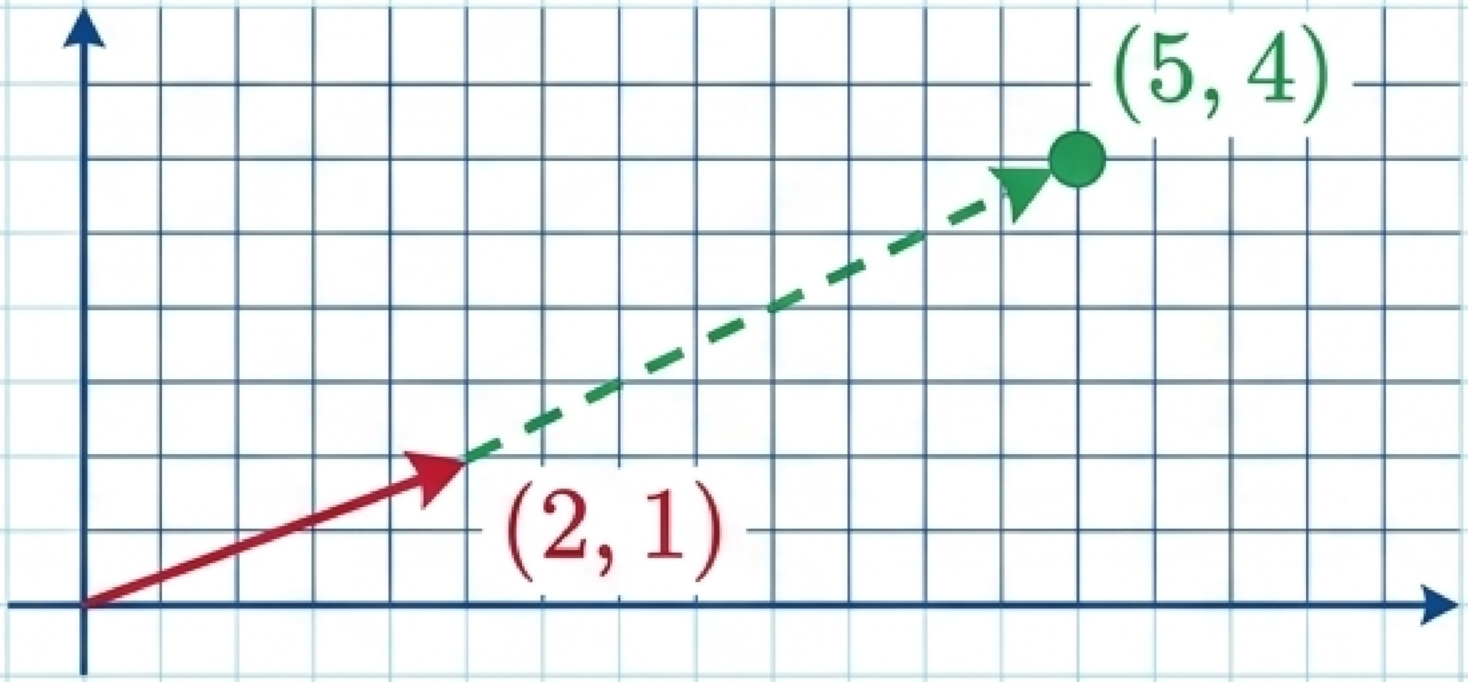
The Vector

$$(x, y)$$

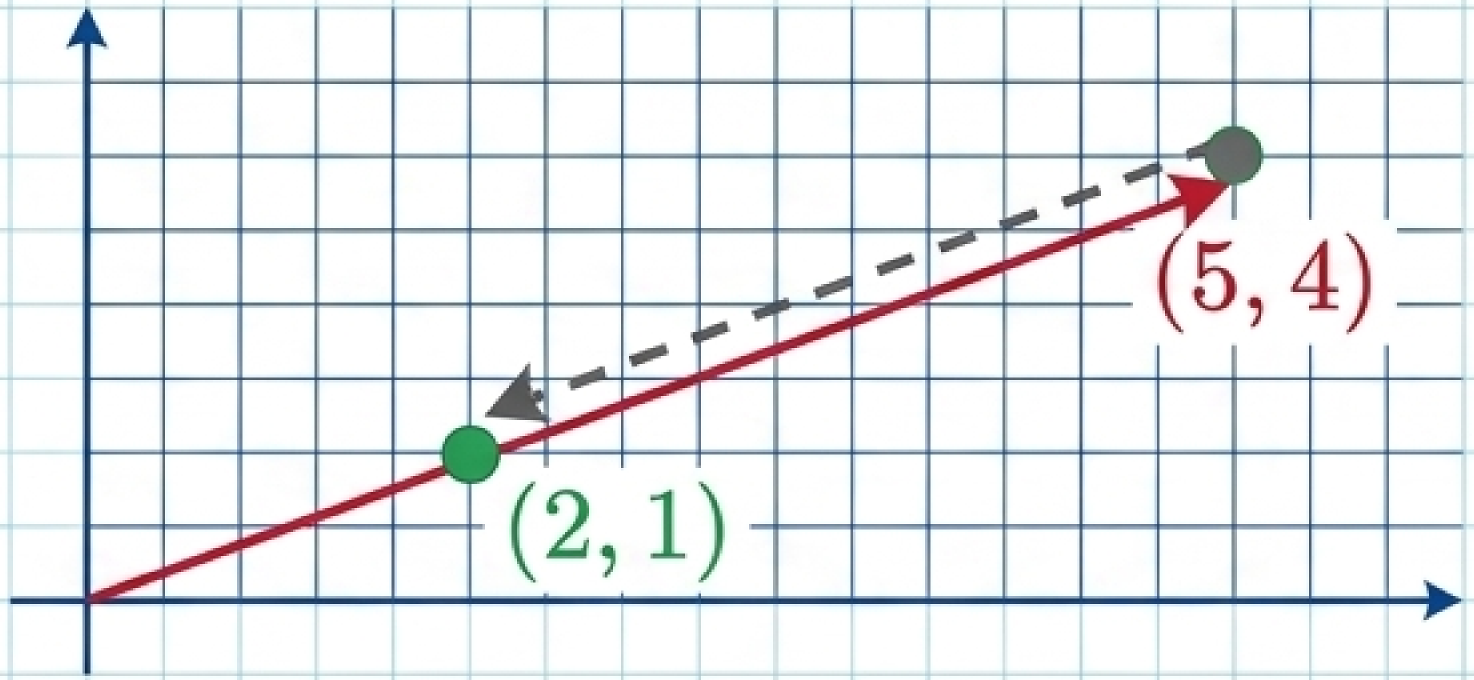
Magnitude AND Direction.

Real-World Example:
Distance traveled towards the right.

ML Application: Representing a single data point in a multidimensional geometric space.



$$\begin{bmatrix}2\\1\end{bmatrix}+3=\begin{bmatrix}5\\4\end{bmatrix}$$

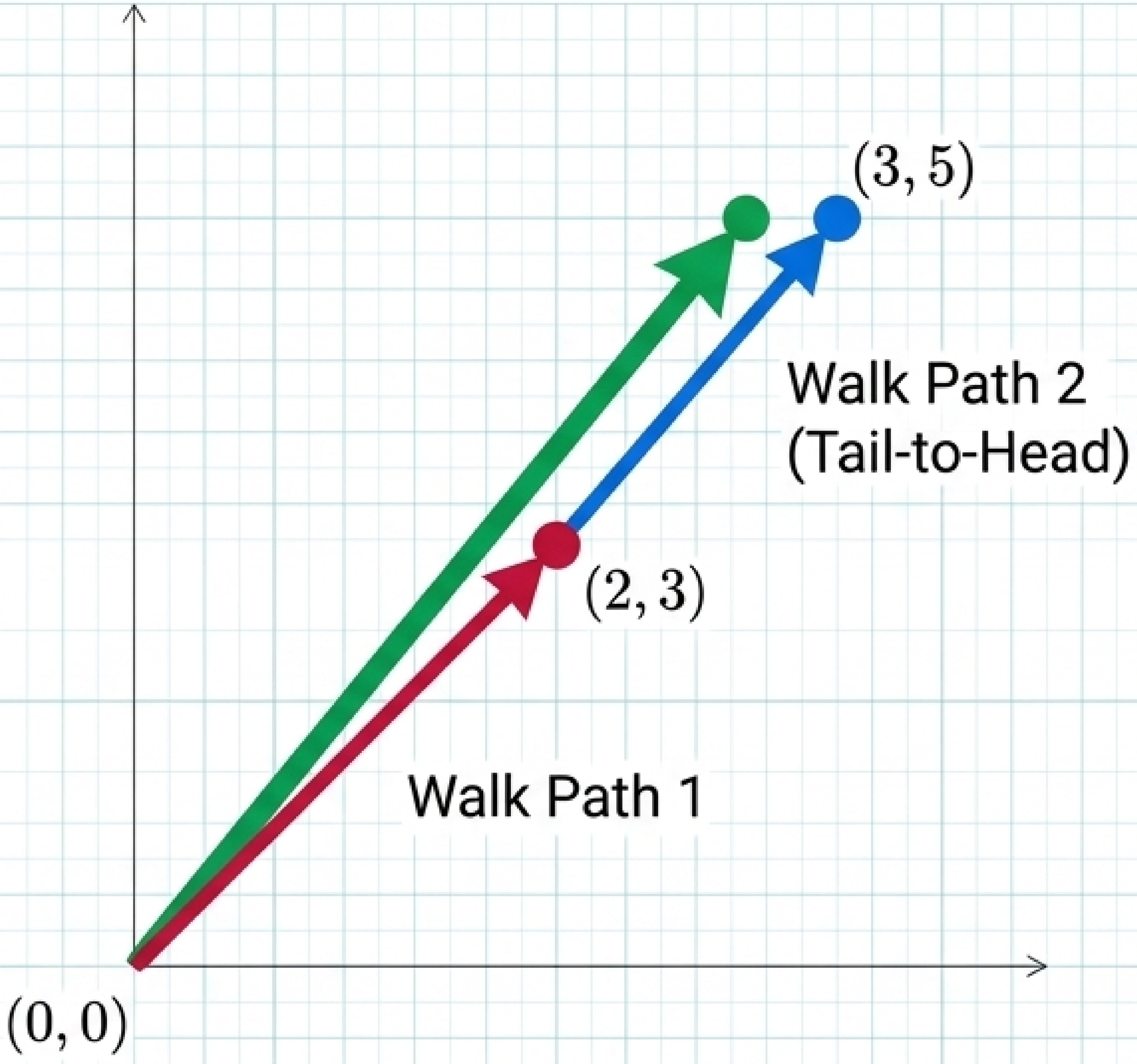


$$[5,4]-3=[2,1]$$

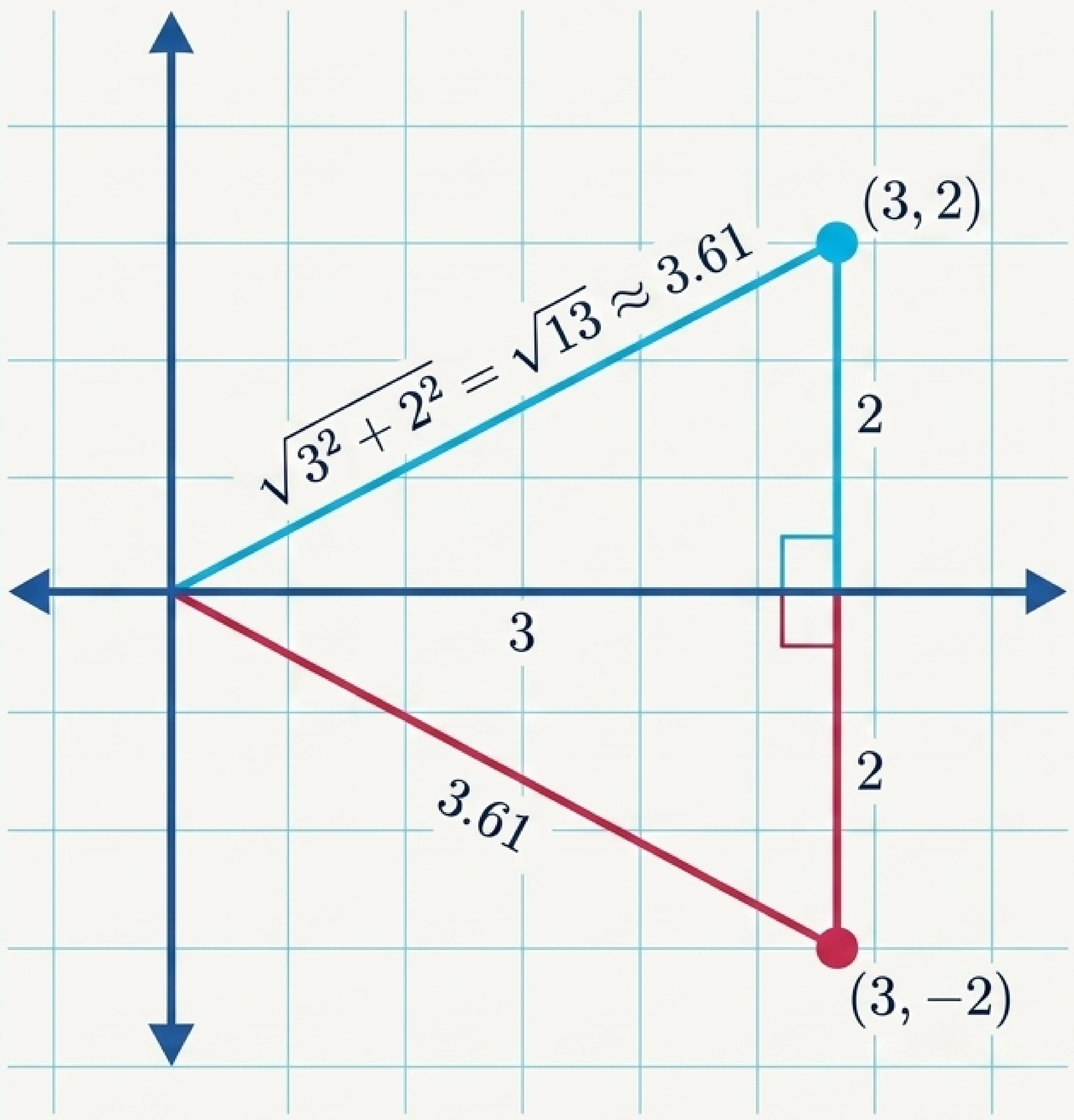
The ML Context

Mean Centering: When we subtract a scalar average from every point, we physically shift the entire dataset to center perfectly around zero.

$$V[2,3] + A[1,2] = [3,5]$$



The **Tail-to-Head Rule:** Adding vectors requires matching dimensions. Geometrically, you walk the path of the first vector, then place the tail of the second vector at your current position and walk its path.



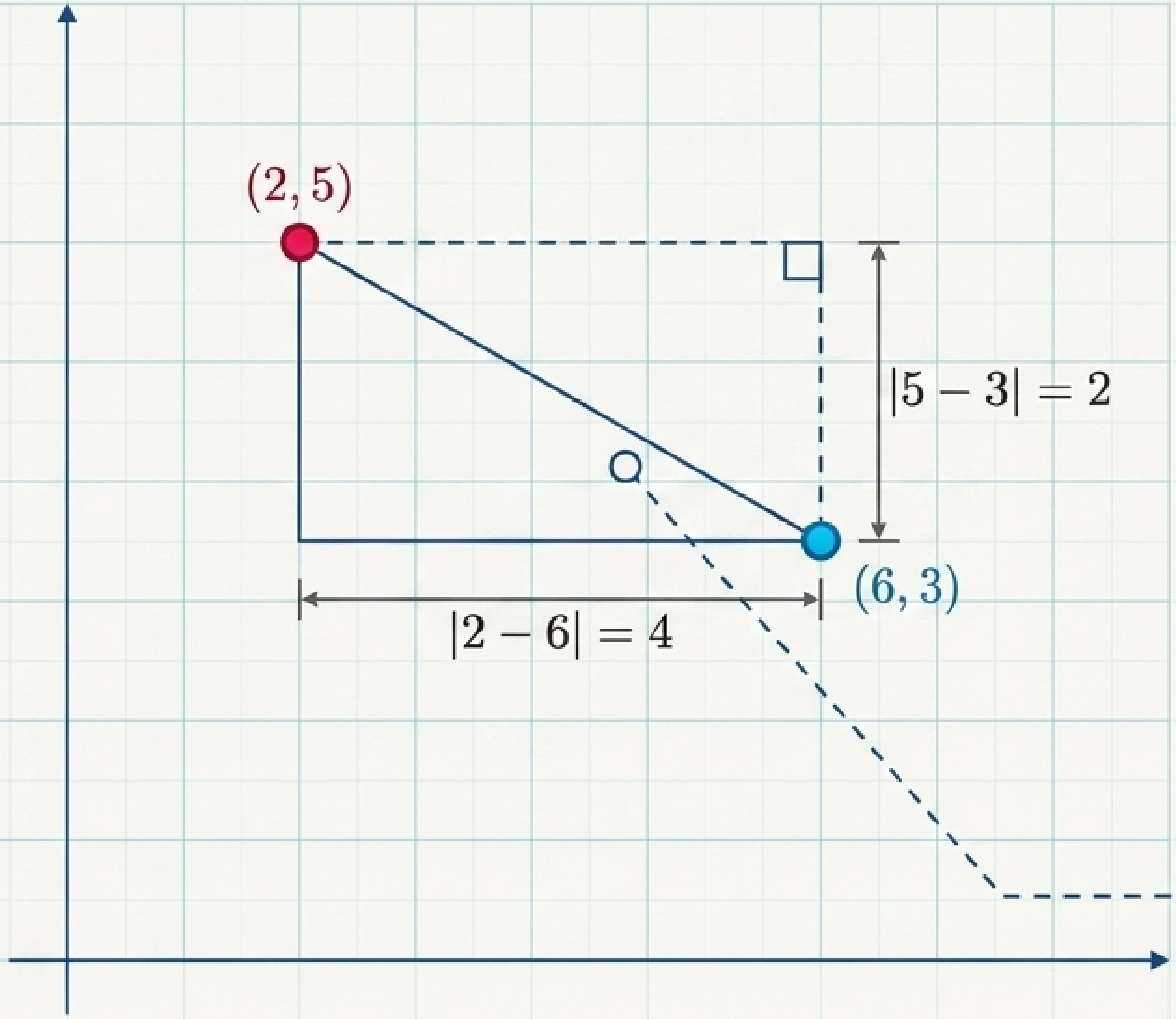
The ML Translation

Feature Importance

A vector with a massive distance from the origin has a mathematically stronger value in space.

L2 Regularization

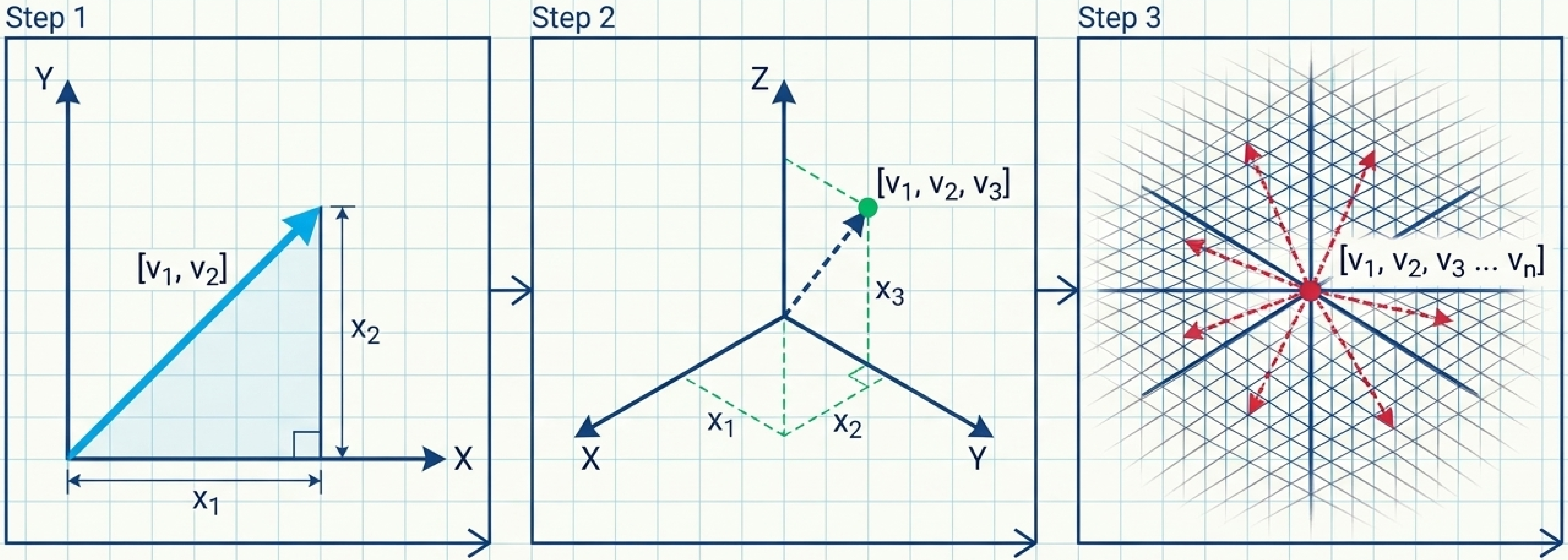
Overfitting often means weights (W) are too far from the origin. The magnitude formula calculates their size so the model can penalize and shrink them.



$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$Distance = \sqrt{(-4)^2 + (-2)^2} = \sqrt{20} \approx 4.47$$

Error Measurement: Calculating the difference between a predicted value and an actual value is literally measuring the spatial Euclidean distance between those two points.



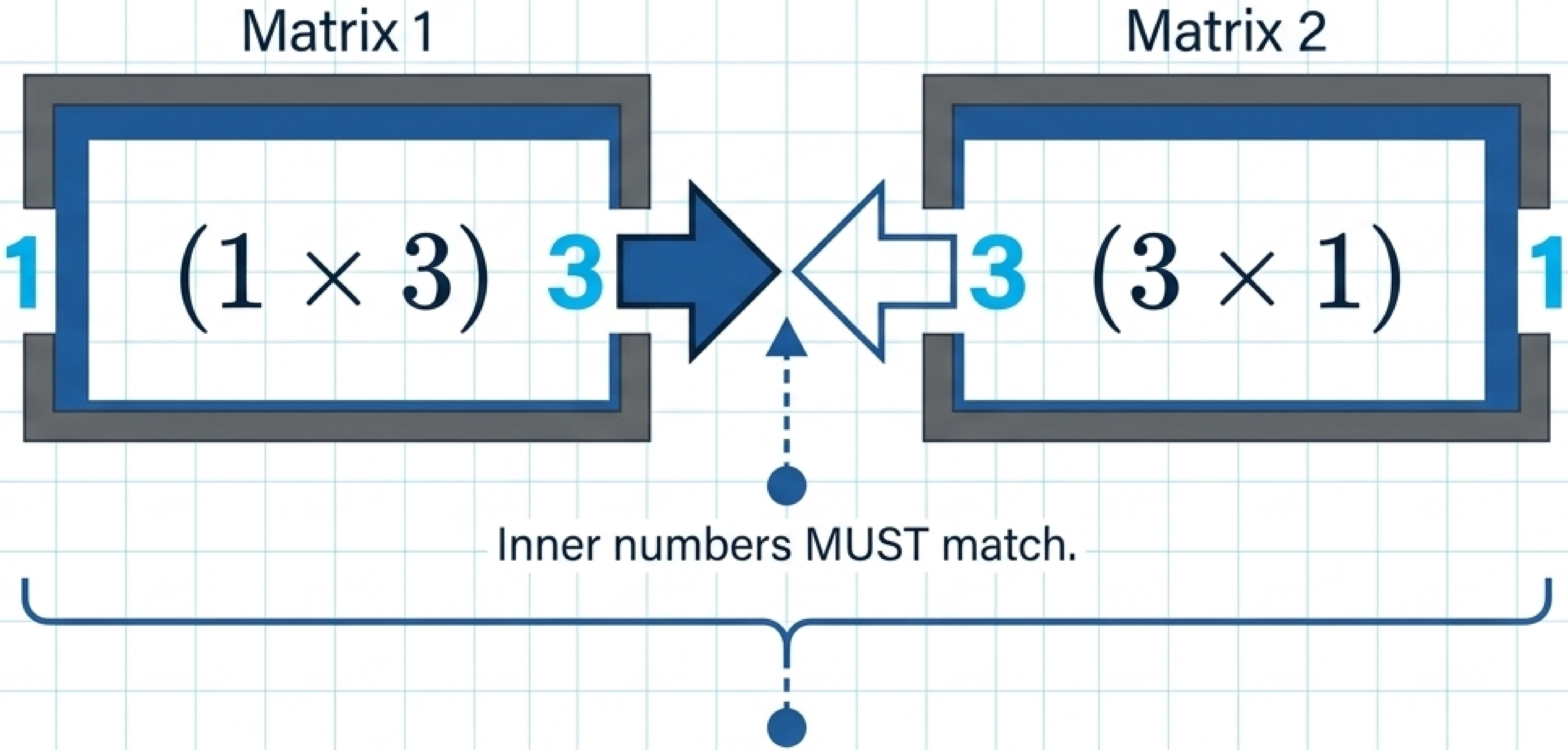
$$\sqrt{x_1^2 + x_2^2}$$

$$\sqrt{x_1^2 + x_2^2 + x_3^2}$$

$$\sqrt{x_1^2 + x_2^2 + x_3^2 \ldots + x_n^2}$$

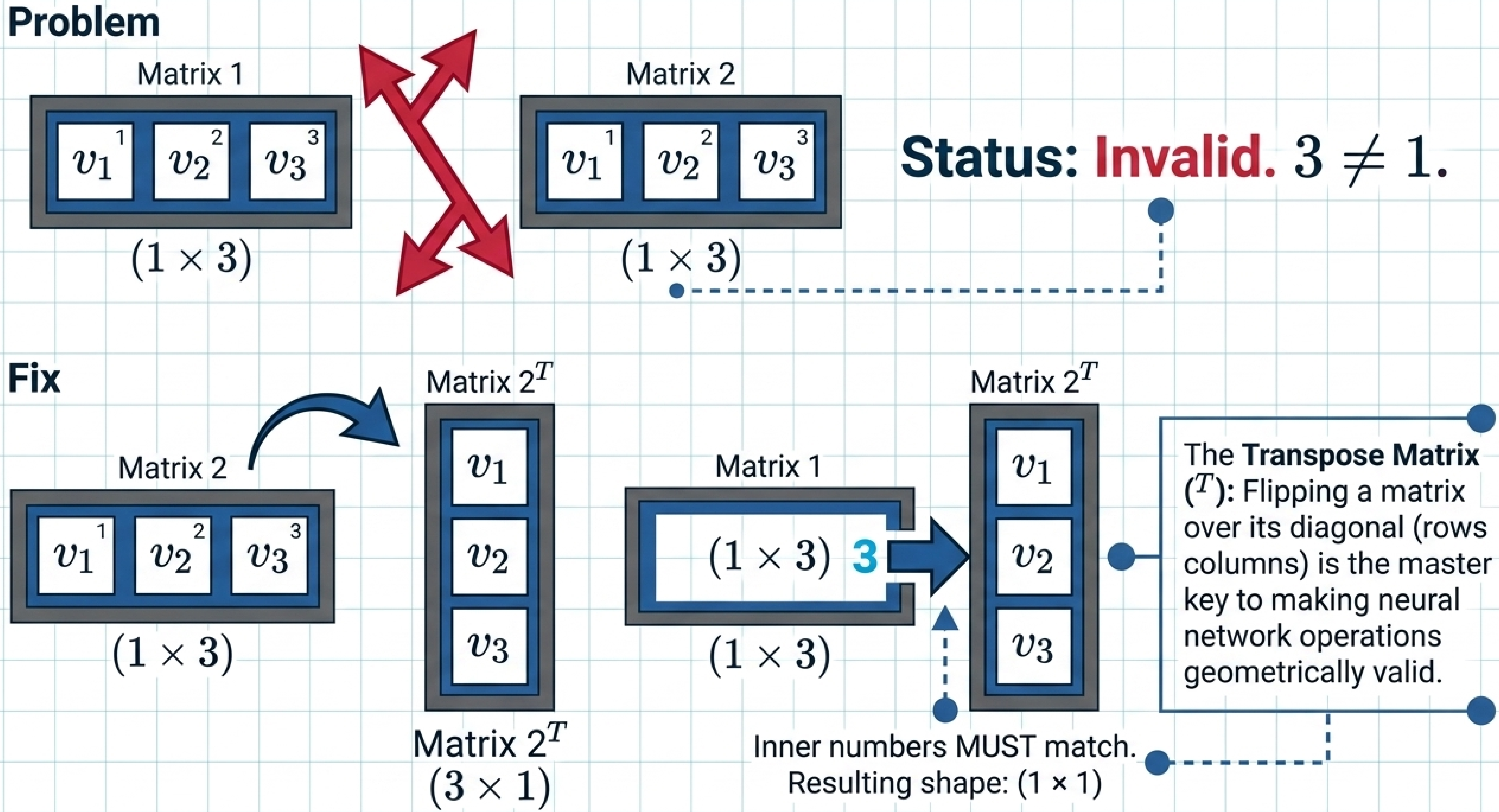
The Mathematical Beauty: Even though we cannot visualize a 100-dimensional dataset, the geometric logic scales perfectly. We just keep adding dimensions to the Pythagorean formula.

Matrix Multiplication: The Golden Rule of Shapes. For an algorithm to combine data layers, the geometric dimensions must perfectly align.



Inner numbers MUST match.

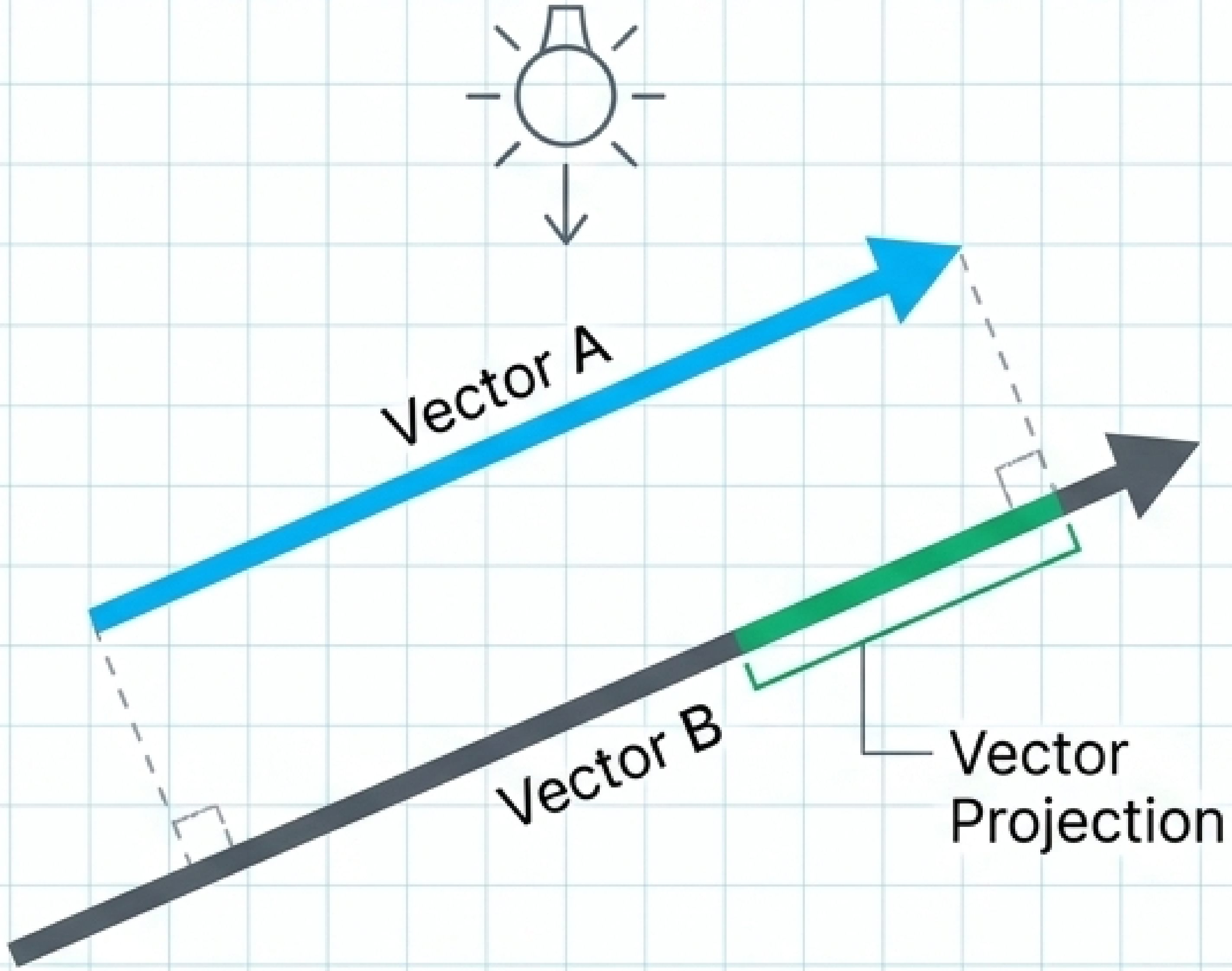
Outer numbers dictate the shape of the new matrix (1 × 1).



Problem
Matrix 1
v_1 1 v_2 2 v_3 3
(1×3)
Matrix 2
v_1 1 v_2 2 v_3 3
(1×3)
Status: Invalid. $3 \neq 1$.
Fix
Matrix 2
v_1 1 v_2 2 v_3 3
(1×3)
Matrix 2^T
v_1 v_2 v_3
Matrix 2^T
(3×1)
Matrix 1
(1×3) 3
(1×3)
Matrix 2^T
v_1 v_2 v_3
Inner numbers MUST match.
Resulting shape: (1×1)
The **Transpose Matrix** (^{T}): Flipping a matrix over its diagonal (rows columns) is the master key to making neural network operations geometrically valid.

NotebookLM

VECTOR PROJECTION & DOT PRODUCT



Equation Box

$$\mathbf{A} \cdot \mathbf{B} = (x_1 \cdot y_1) + (x_2 \cdot y_2) + (x_3 \cdot y_3)$$

ML Applications

Cosine Similarity: Measures the exact angular distance between vectors to find data similarity.

Dimensionality Reduction (PCA): Uses projections to compress massive datasets while preserving the most important geometric information.

Neural Networks: Returns a single scalar number from two vectors.

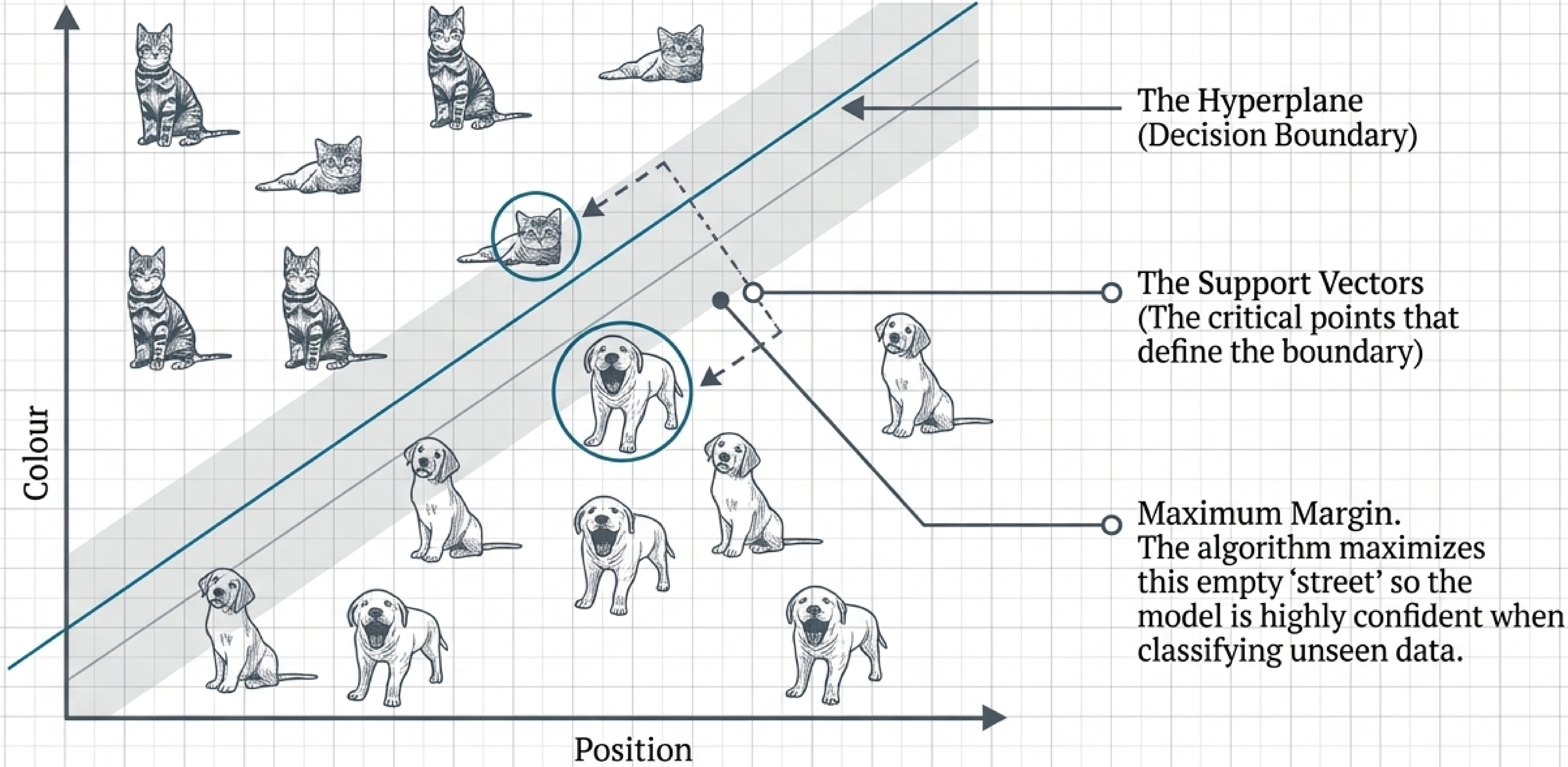
$$y = w_1x_1 + w_2x_2 + w_3x_3 + b$$

Simplified via **Dot Product** & **Transposition**

$$y = X + bW^T$$

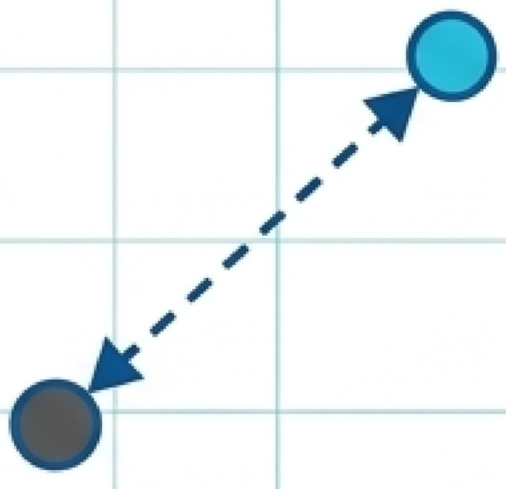
The fundamental calculation inside Neural Networks is just a **Dot Product**: multiplying an input vector (X) by a **weight vector** (W), transposed to ensure the matrix shapes fit perfectly.

LINEAR CLASSIFICATION & MARGIN OPTIMIZATION



The Blueprint Synthesized: Math to Algorithm

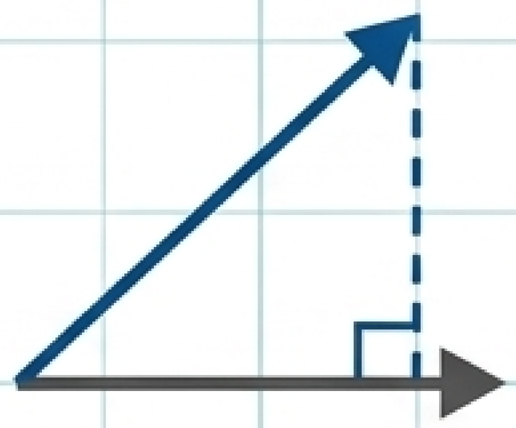
Navigation



Math Core:
Euclidean Distance & Vector Proximity

ML Algorithm: K-Nearest Neighbors (KNN)

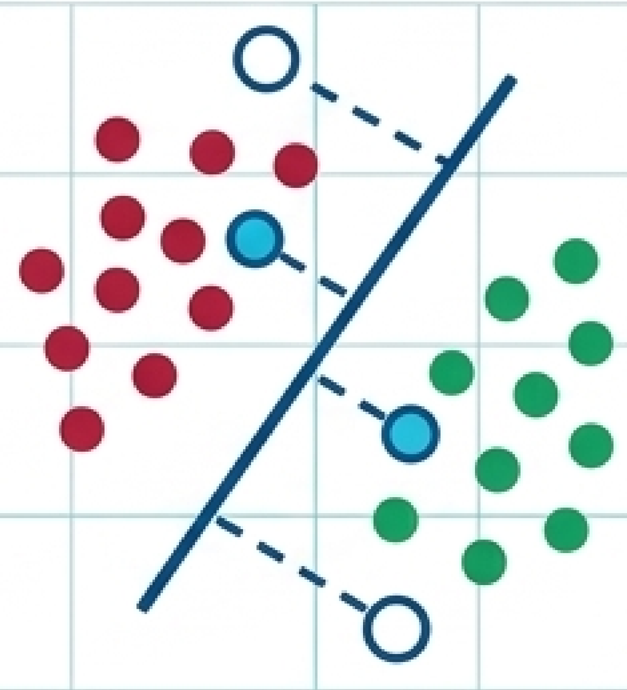
The Engine



Math Core:
The Dot Product & Matrix Multiplication

ML Algorithm: Linear Layers / Neural Networks

The Boundaries



Math Core:
Maximizing Geometric Margins

ML Algorithm: Support Vector Machines (SVM)

The math is the **engine**; the algorithm is simply the vehicle.